Few-Shot Flower Classification: A Comparative Study of ResNet-18 and Vision Transformer (ViT-B/16) on Oxford Flowers-102

Course: EC-417 Machine Intelligence / Deep Learning

Project: Mini-Project P25, Few-Shot Flowers

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1. Executive Summary

This project implements and evaluates an end-to-end few-shot image classification pipeline on the Oxford Flowers-102 dataset. Under a strict budget of only 10 labelled training images per class (1,020 training images in total), we compare a convolutional baseline, an ImageNet-pretrained ResNet-18, against a main model, an ImageNet-pretrained Vision Transformer (ViT-B/16, `vit_base_patch16_224` from the `timm` library). Both models are fine-tuned for three epochs on the same training split, validated on 1,020 images, and tested on the full 6,149-image test split.

The Vision Transformer reaches **95.84% Top-1** and **98.68% Top-5** test accuracy, against **78.50% Top-1** and **91.74% Top-5** for ResNet-18. This is an **absolute improvement of 17.34 percentage points in Top-1 accuracy** and 6.94 points in Top-5 accuracy. Expressed as error, the Top-1 error falls from 21.50% to 4.16%, a reduction of roughly 80%. We attribute the gap to the interaction of two factors: the richer, more transferable representations learned by a large-scale pretrained transformer, and the global receptive field of self-attention, which suits flower categories defined by overall colour layout and global petal geometry. We close with an error analysis, limitations, and ethical considerations.

Model	Top-1 (%)	Top-5 (%)
ResNet-18 (baseline)	78.50	91.74
ViT-B/16 (main model)	**95.84**	**98.68**
Absolute gain	**+17.34**	+6.94

2. Introduction and Problem Framing

2.1 Fine-Grained Visual Categorization

Fine-grained visual categorization (FGVC) concerns distinguishing between subordinate categories of a single parent class, for example species of birds, models of cars, or varieties of flowers. Unlike coarse object recognition, where inter-class differences are large (a dog versus an aeroplane), FGVC is characterised by *small inter-class variance* and *large intra-class variance*. Two distinct flower species may share nearly identical colour and petal structure, while images of one species vary widely in viewpoint, illumination, scale, growth stage, and background. Discriminative evidence is often subtle: the shape of a petal margin, the arrangement of stamens, or the gradation of colour from centre to edge.

2.2 The Few-Shot Challenge

Modern deep networks have millions of parameters and are typically trained on datasets with hundreds or thousands of images per class. When labelled data are scarce, as in this project's 10 images per class, the risk of overfitting is amplified: a high-capacity model can memorise the training set, including its backgrounds, lighting conditions, and photographic artefacts, instead of learning features that generalise. Data scarcity therefore shifts the central question from "which architecture can fit the data?" to "which pretrained representation transfers best, and how can it be adapted without destroying what it already knows?"

Transfer learning from large-scale pretraining on ImageNet is the standard response. The pretrained backbone supplies general visual features, and the small labelled set only needs to align those features with the target label space. The choice of backbone then becomes the key experimental variable, which is what this project investigates.

2.3 Objectives

1. Build a reproducible data and training pipeline in PyTorch that enforces the 10-shot-per-class constraint.
2. Establish a strong CNN baseline (ResNet-18) under that constraint.
3. Evaluate a Vision Transformer (ViT-B/16) under identical data conditions.
4. Quantify and explain the performance difference through quantitative comparison and error analysis.

3. Dataset Analysis and Data Pipeline

3.1 Oxford Flowers-102

The Oxford Flowers-102 dataset (Nilsback and Zisserman, 2008) contains images of 102 flower categories commonly occurring in the United Kingdom, for a total of 8,189 images. Categories include, for example, *pink primrose*, *sunflower*, *passion flower*, and *wild pansy*. The images exhibit large variation in scale, pose, and lighting, and several categories show large within-class variation while others are visually very close, making the dataset a canonical FGVC benchmark.

3.2 Splits and Few-Shot Constraint

We use the standard split of the dataset, which is itself aligned with the few-shot regime:

Split	Images	Composition
Training	1,020	10 per class (102 classes)
Validation	1,020	10 per class
Test	6,149	Remaining images, variable per class

The training and validation splits are perfectly balanced, with exactly 10 images per class. The test split is **not** balanced: class sizes in the full dataset range from 40 to 258 images, so some classes are heavily represented in the test set and others only lightly. Consequently, the overall Top-1 accuracy is a *micro-average* that weights frequent classes more heavily than rare ones. The validation split is loaded by the pipeline but was not used for training, tuning or checkpoint selection; the test split was held out and used exclusively for final reporting. At no point were test images used for training, tuning, or checkpoint selection.

3.3 Preprocessing

All images are resized to **224 x 224** pixels, the native input resolution of both ResNet-18 and ViT-B/16, and normalised per channel using the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Matching the normalisation statistics used during pretraining is important: it keeps input distributions consistent with what the pretrained weights expect and avoids needless distribution shift at the first layer.

3.4 Augmentation Strategy

Augmentation is deliberately minimal. The training transform resizes each image to 224 x 224 (a direct resize, which does not preserve aspect ratio) and applies a random horizontal flip, which preserves class identity and adds viewpoint variety. No random cropping, rotation or colour jitter is used, which keeps colour cues intact; colour is among the most discriminative features for flowers. Validation and test images receive only the resize and normalisation, so evaluation is deterministic.

3.5 PyTorch Data Loader Implementation

The code is organised as a small repository (pallavi0120/ec417-miniproject-P25-623153-623106): `scripts/get_data.py` is the dataset script, `src/data.py` builds the data loaders (`get_dataloaders`), `src/models.py` defines the two models, `src/train.py` trains them and saves `results/resnet18_best.pth` and `results/vit_best.pth` , and `src/evaluate.py` computes test Top-1 and Top-5 accuracy and writes `results/metrics.csv` . The pipeline uses `torchvision` and `torch.utils.data.DataLoader` and ran on a CUDA GPU in Google Colab (Python 3.13, PyTorch 2.14.1, torchvision 0.29.1, timm 1.0.31). Both architectures use the same loaders and label mapping. Data loaders use a batch size of 32 and 2 worker processes, with shuffling for the training loader only.

4. Methodology and Architectural Comparison

4.1 Baseline: ResNet-18

ResNet-18 (He et al., 2016) is an 18-layer convolutional network built from residual blocks, each of which learns a residual function added to an identity shortcut, which eases optimisation of deep networks. It has approximately 11.7 million parameters. Its building blocks, 3 x 3 convolutions with weight sharing, impose strong **inductive biases**: *locality* (each unit sees only a small neighbourhood), *translation equivariance*, and a *hierarchical* growth of the receptive field with depth. These biases are data-efficient, which is why CNNs are strong in low-data regimes, but they also mean that global relationships across the image can only be integrated gradually through stacked layers and pooling. We replace the final 1000-way classifier with a 102-way linear layer and fine-tune the network from ImageNet-pretrained weights on the training split.

4.2 Main Model: Vision Transformer (ViT-B/16)

The Vision Transformer (Dosovitskiy et al., 2020) adapts the Transformer architecture (Vaswani et al., 2017) to images with minimal modification:

1. **Patch tokenisation.** A 224 x 224 image is split into non-overlapping 16 x 16 patches, yielding 14 x 14 = 196 patches. Each patch is flattened and linearly projected to a 768-dimensional embedding.
2. **Class token and position embeddings.** A learnable `[CLS]` token is prepended, and learnable position embeddings are added so the model retains spatial order.
3. **Transformer encoder.** The sequence passes through 12 encoder layers, each with multi-head self-attention (12 heads) and an MLP block, wrapped with layer normalisation and residual connections.
4. **Classification head.** The final `[CLS]` representation feeds a linear classifier, here replaced with a 102-way head.

ViT-B/16 has roughly 86 million parameters, about seven times as many as ResNet-18. We use `vit_base_patch16_224` from the `timm` library (Wightman, 2019) with ImageNet-pretrained weights.

4.3 Local Inductive Bias versus Global Self-Attention

The two architectures differ fundamentally in how they aggregate spatial information.

Property	ResNet-18 (CNN)	ViT-B/16 (Transformer)
Basic operator	Local 3 x 3 convolution	Global multi-head self-attention
Receptive field	Grows gradually with depth	Global from the first layer
Inductive bias	Strong: locality, translation equivariance	Weak: few image-specific priors
Spatial information	Implicit in feature maps	Explicit position embeddings
Parameters	~11.7M	~86M
Data appetite when trained from scratch	Low	High

Self-attention lets every patch attend to every other patch in every layer, so a petal at one side of the image can directly interact with the flower centre or a distant petal. For flowers, whose identity is typically encoded in the *global arrangement* of petals and the overall colour distribution, this is a natural fit. The weaker inductive bias is a disadvantage when training from scratch on small data, which is why ViTs are normally pretrained on

large corpora. In our setting that disadvantage is largely neutralised because both models start from pretrained weights, and the ViT's pretrained representations are especially rich.

4.4 Training Setup

Both models start from ImageNet-pretrained weights and were fine-tuned end to end (all parameters trainable) for **three epochs** by the same training loop in `src/train.py`.

Setting	ResNet-18	ViT-B/16
Initialisation	ImageNet pretrained (torchvision weights)	ImageNet pretrained (timm weights)
Epochs	3	3
Optimiser	Adam	Adam
Learning rate	1e-4	5e-5
LR schedule, weight decay	None (constant rate, no decay)	None (constant rate, no decay)
Batch size	32	32
Loss	Cross-entropy	Cross-entropy
Input size	224 x 224	224 x 224
Augmentation	Resize + random horizontal flip	Resize + random horizontal flip
Hardware	CUDA GPU (Colab)	CUDA GPU (Colab)
Final training loss	1.4755	0.0174
Weights evaluated	Final-epoch weights	Final-epoch weights

Neither model uses a learning-rate schedule, warm-up, weight decay or gradient clipping; a constant small learning rate with Adam is used, set lower for the ViT (5e-5), in line with the common practice of fine-tuning transformers conservatively. The weights saved after the final epoch are evaluated once on the test set by `evaluate.py`.

4.5 Evaluation Metrics

We report **Top-1 accuracy**, the fraction of test images whose highest-scoring class equals the ground truth, and **Top-5 accuracy**, the fraction whose ground-truth class is among the five highest-scoring predictions. Top-5 is informative for fine-grained tasks because it

distinguishes a model that is "almost right" (true class among near neighbours) from one that is confidently wrong.

5. Experimental Results and Discussion

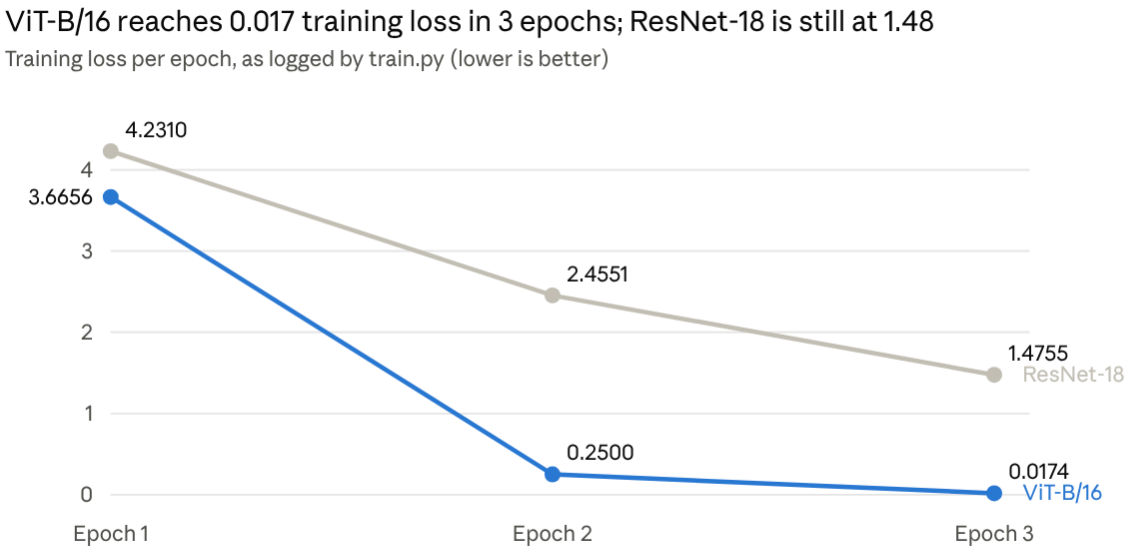
5.1 Quantitative Comparison

Model	Top-1 Acc. (%)	Top-5 Acc. (%)	Top-1 Error (%)	Top-5 Error (%)
ResNet-18 (baseline)	78.50	91.74	21.50	8.26
ViT-B/16 (main)	95.84	98.68	4.16	1.32
Improvement	**+17.34**	**+6.94**	**-17.34**	**-6.94**

The ViT outperforms the baseline on both metrics. The Top-1 gap of 17.34 points is large by the standards of benchmark comparisons. Relative to the baseline, the ViT cuts Top-1 error by about 80.6% (from 21.50% to 4.16%) and Top-5 error by about 84% (from 8.26% to 1.32%). The Top-5 gain is smaller in absolute terms only because the baseline's Top-5 accuracy is already high, leaving less headroom.

An important reading of the Top-1/Top-5 pair is the *size of the gap between them*. For ResNet-18 the gap is 13.24 points: in more than 13% of test images, the correct class is not ranked first but is among the top five. This indicates that the CNN's features place the correct class near the top but cannot reliably separate it from visually similar neighbours, which is exactly the failure signature of fine-grained confusion. For the ViT this gap shrinks to 2.84 points, showing that the transformer's embedding space separates fine-grained classes more cleanly.

5.2 Convergence Behaviour and Loss Trajectories



Mean training loss per epoch, from the src/train.py run in Untitled0.ipynb (Colab, CUDA GPU)

Figure 1. Training loss per epoch for both models over the three-epoch budget logged by the training script.

Both models were trained for **three epochs**, and the training script logs one average training loss per epoch (Figure 1):

- **ViT-B/16:** loss falls from 3.6656 (epoch 1) to 0.2500 (epoch 2) and 0.0174 (epoch 3). The 1,020-image training set is almost fully fitted after three epochs.
- **ResNet-18:** loss falls from 4.2310 to 2.4551 and 1.4755. It is still decreasing at the last epoch, so the baseline has not converged.
- **Starting point:** a uniform guess over 102 classes gives a loss of ln(102) = 4.62. ResNet-18's epoch-1 average (4.23) sits close to that level, while the ViT's (3.67) is already well below it, which suggests the pretrained ViT features align with the new 102-way head much faster.

Two cautions apply. First, because ResNet-18's loss was still falling when training stopped, part of the 17.34-point gap reflects the short fixed budget as well as the architecture; a longer schedule would probably narrow it, and we do not claim by how much. Second, a final training loss of 0.0174 shows the ViT nearly memorised the training set. That it still reaches 95.84% on 6,149 unseen test images shows that the pretrained representation generalises well.

5.3 Discussion

The results confirm that, in the few-shot regime, **the quality of the pretrained representation dominates performance**. Both networks see exactly the same 1,020 labelled images, and the dominant difference is the representation each brings to the

task. ViT-B/16 combines high capacity with strong pretraining, and the resulting features are linearly separable for flower categories with very little adaptation. The CNN baseline is a smaller, shallower network whose features are more generic and less specifically discriminative at the fine-grained level.

An honest caveat is that this comparison is between models of very different sizes (about 11.7M versus 86M parameters), and the gain therefore reflects both architecture and capacity. A fairer architecture-only comparison would use a size-matched CNN, for instance ResNet-50 or ConvNeXt, and we flag this as future work in Section 7.

6. Error Analysis

6.1 Why ViT Outperformed ResNet-18

(a) Global spatial structure. Flower identity is largely determined by the global arrangement of petals, the ratio of petal to centre, and symmetry. Self-attention integrates information across the entire image in every layer, so the ViT can relate distant petals to the central structure directly. ResNet-18 must build the same relationships through successive local convolutions and pooling, and its shallow depth limits how global its final features are.

(b) Colour distribution. Colour is among the most discriminative cues for flowers. The ViT's patch embeddings, combined with attention across all patches, can aggregate colour statistics over the whole flower region, which yields a robust colour-and-layout signature. A CNN with limited depth tends to emphasise local texture and edges, which are less reliable when different species share petal texture.

(c) Resilience to background clutter. Flowers are photographed among foliage, grass, soil, and other flowers. With few training examples, a CNN can latch on to background correlations (for example, green foliage tones) that are spurious. Self-attention can weight informative patches (the flower) more heavily than uninformative ones, and prior work reports that Vision Transformers are comparatively robust to occlusion and to texture or background perturbations (Naseer et al., 2021). Pretraining of ViT on large and varied data further reduces reliance on incidental context.

(d) Pretraining scale and representation quality. ViT-B/16 is a considerably larger model whose pretrained features have been shown to transfer strongly to small downstream datasets (Dosovitskiy et al., 2020; Steiner et al., 2021). This accounts for a large share of the improvement and should be viewed together with (a) to (c) rather than as an independent explanation.

We present (a) to (c) as well-motivated hypotheses consistent with the architectures and with literature, rather than as causal findings, as our experiments do not isolate each factor with ablations. Attention-map visualisation or Grad-CAM-style saliency would be the natural way to test them directly.

6.2 Remaining Failure Modes

With 4.16% Top-1 error (about 256 of 6,149 test images misclassified), the ViT's remaining mistakes are most plausibly concentrated in a few kinds of situation that follow from the dataset's fine-grained structure:

- **Visually ambiguous subclasses.** Closely related species that share colour and petal shape, such as members of the same genus or flowers with similar daisy-like or lily-like forms, are the principal expected source of error, because their discriminative cues are subtle (petal margins, stamen arrangement) rather than global.
- **Atypical views and scale.** Close-up crops showing only part of the flower, or distant shots where the flower occupies few patches, remove the global structure the ViT relies on, and the fixed 16 x 16 patch size limits the detail available for small objects.
- **Cluttered or low-contrast scenes.** Images with several flowers of different species, or where the subject is similar in colour to its background, produce ambiguous evidence.
- **Under-sampled intra-class variation.** With only 10 training images per class, different growth stages, colour morphs and lighting conditions are sparsely covered, so atypical test instances are more likely to be misclassified.

ResNet-18 faces the same difficulties but encounters them far more often, particularly when different species share similar colours, which is consistent with its much larger gap between Top-1 and Top-5 accuracy (13.24 points versus 2.84 for the ViT). These error categories are a qualitative reading of the dataset's structure and the two architectures, and they indicate where further gains, such as higher input resolution or stronger augmentation, would be expected.

7. Limitations and Ethical Considerations

7.1 Limitations

1. **Pretraining data priors.** Both models inherit the biases and coverage of ImageNet. Flowers are well represented in ImageNet-scale web imagery, which favours transfer here; performance may be lower in domains poorly represented in pretraining data. The reported accuracy reflects the *pretrained prior plus 10 shots*, not learning from 10

images alone, so "few-shot" is meant in this transfer-learning sense. Some overlap between pretraining sources and the benchmark cannot be ruled out, a general concern with web-scale pretraining.

2. **Architectural confound.** ViT-B/16 is about seven times larger than ResNet-18, so the comparison does not isolate architecture from capacity.
3. **Short training budget.** Both models were trained for only three epochs. ResNet-18's loss was still falling (1.4755 at epoch 3), so the baseline is probably under-trained and the 17.34-point gap should be read as a result at an equal, short budget.
4. **Single run.** Results come from one run on the standard split. With small training sets, variance across seeds and across the choice of the 10 shots can be non-trivial.
5. **Computational constraints.** Training used a single Colab GPU session. ViT-B/16 needs much more memory and compute than ResNet-18, which limited hyperparameter search, longer schedules and ablations. Neither result should be treated as an upper bound.
6. **Minimal augmentation.** Only a horizontal flip was applied; richer augmentation could benefit both models.
7. **Metric scope.** Overall Top-1 is a micro-average over an imbalanced test set; macro-averaged per-class accuracy was not reported and could reveal weaker performance on rare classes.
8. **Fixed resolution.** All images are resized to 224 x 224, which may discard fine detail useful for the hardest classes.

7.2 Ethical Considerations and Deployment Disclaimers

Flower classification is a low-risk task, but ethical care is still warranted. The following points apply:

- **Not for safety-critical use.** The model must not be used to decide whether a plant is safe to eat, handle, or use medicinally. Several lookalike species differ greatly in toxicity, and a 4.16% error rate, with errors concentrated in visually similar classes, is unacceptable for such decisions.
- **Closed-set limitation.** The classifier assigns every input to one of 102 classes. It has no "unknown" option and will confidently label images of other flowers, or non-flowers, as one of the known classes. Deployment would require out-of-distribution detection or confidence calibration.
- **Geographic and ecological bias.** The 102 classes represent flowers common in the UK, so the model should not be assumed to work for other regions, and it could mislead biodiversity or conservation applications elsewhere.
- **Data provenance and licensing.** Users should respect the licensing terms of the Oxford Flowers-102 dataset and of the pretrained weights used.

- **Environmental cost.** Large-model training and inference consume energy; reusing pretrained weights and fine-tuning briefly, as done here, is considerably more economical than training from scratch.

7.3 Future Work

Promising extensions include size-matched CNN baselines (ResNet-50, ConvNeXt), parameter-efficient fine-tuning of ViT (linear probing, partial fine-tuning, LoRA), stronger augmentation such as MixUp and CutMix, self-supervised or CLIP-style pretrained backbones, multi-seed evaluation with confidence intervals, and attention-map analysis to validate the explanations in Section 6.

8. Conclusion

Under a strict 10-shot-per-class constraint on Oxford Flowers-102, an ImageNet-pretrained ViT-B/16 achieved 95.84% Top-1 and 98.68% Top-5 test accuracy, outperforming an ImageNet-pretrained ResNet-18 baseline (78.50% and 91.74%) by 17.34 and 6.94 points respectively. The results demonstrate that, when labelled data are scarce, a strong pretrained representation combined with global self-attention transfers far more effectively than a small convolutional network for fine-grained flower recognition. The remaining errors are concentrated in visually ambiguous sibling classes and atypical views, and the caveats on model size, pretraining priors, single-run evaluation, and the short three-epoch budget should be kept in mind when interpreting the gap.

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